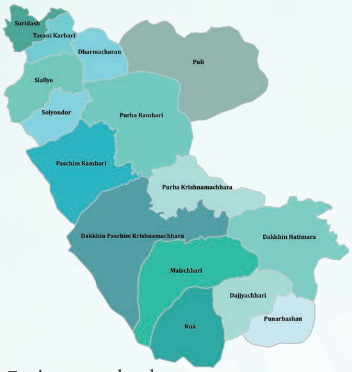
Gilachhari Watershed

R-114

Figure 1:
Para (village) boundary of Gilachhari watershed



Source:
Center for Environmental and Geographic Information Services (CEGIS)

Disclaimer: Boundaries are not necessarily authoritative

The Gilachhari sub-watershed (R-114), located in the Ghilachari and Burighat Unions, 71 no Choto Maurum Mauza and 78 no Bagachari Mauza of the Naniarchar Upazila of Rangamati District, encompasses 15 paras (figure 1) and covers approximately 1,897 hectares of land. It is home to over 5,000 people, with communities consisting of the Chakma, Marma, and Tripura ethnic groups. The overall literacy rate is approximately 70%, yet opportunities for higher education are very limited. Farming is the main occupation. The average monthly income is 11,230 taka ($94), which is relatively low compared to the national average. About 99% of the houses are katcha and almost 93% of households use solar electricity systems. There is a noticeable scarcity of drinking water and a prevalent incidence of waterborne diseases.

Topography

Only 13% area lies below 40-meter (m) elevation (figure 2). The valley situated in the centre of the sub-watershed has the lowest elevation in the region. Most of the study area lies between 40–60m (29%) and 61–80m (31%) elevation. These regions are mostly situated in the North-East region of the watershed, on the east face of the valley. The west face of the valley has comparatively higher elevation. About 20% of the area lies between 81–100m elevation. Only 7% of the area lies above 100m elevation, which is situated in the western region of the watershed.

Figure 2: Topography

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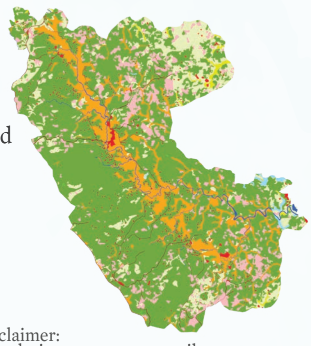
<40 | 60 | 80 | >100

meter Mean Sea Level (mMSL)

Landuse

A total 12 types of land use have been identified within the Gilachhari watershed (figure 3). The main land use is forest (1,867 hectares (ha)), followed by cropland (360 ha), shifting cultivation (264 ha), and herbaceous crop (22 ha).

Figure 3: Land use



- Forest Tree Dominated Area (63%)
- Crop Land (12%)
- Herb/Shrub Dominated Area (12%)
- Rural Settlement (1%)
- Shifting Cultivation (9%)
- Orchard (1%)
- Others (2%)

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Demography

Population

51% | 49%

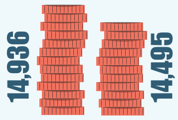
Total 5,034

Literacy

Male 74% | Female 66%

Average literacy rate 70%

Economy



Monthly Income (BDT) | Monthly Expense (BDT)

WASH

- 221 tubewell
- 11687 sanitary latrines
- 139 dug wells
- 6 ring wells
- 32 chharas (creeks)

Poverty

Very Poor	24%
Poor	35%
Middle Class	32%
Better Off	10%

10 | 50 | 100

Production

- Rice 1-2 metric tons per hectare (MT/acre)
- Fisheries 5-6 MT/acre
- 19,950 Livestock

Climatic Hazards

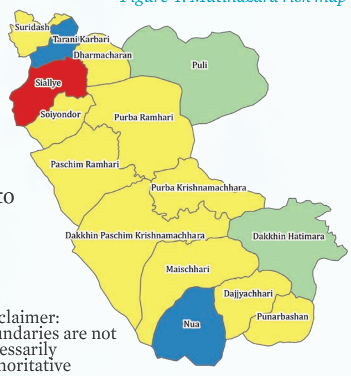
Climate Change-Induced Hazards

Nine climate hazards threaten Gilachhari watershed agriculture, livelihood, ecosystem, and biodiversity. Drought, erratic rainfall, flash floods, heat waves, river floods, landslides, and lightning intensify across most of the paras. Cold waves and tornadoes occur at low to moderate intensity.

Climate Change Risk

Climate change risk assessment identified 1 para as very high-risk hotspot, none as high-risk, 10 as medium-risk, and the remaining paras as low to very low-risk (figure 4).

Figure 4: Mutihazard risk map



Very High	High	Medium	Low	Very Low
1	0	10	2	2

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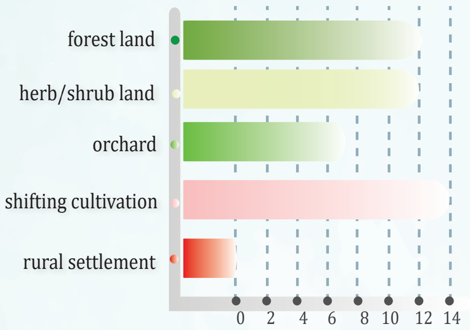
Watershed Degradation

Land Degradation Status

In R-114, land degradation is prominent in -



Causes of Degradation

- Jhum cultivation with short rotation
- Improper crop management
- Overgrazing
- Deforestation & overharvesting of trees
- Lack of vegetation cover
- Climate change
- Overuse of inorganic fertilizers

Degradation Triggering Factors

- Reduction of vegetation cover
- Quality and species composition change/diversity decline/monoculture plantation
- Soil erosion/landslides
- Surface sealing/hard pan development
- Waterlogging
- River bank erosion

Watershed Conservation

Conservation measures focused on enhancing ecological resilience as well as improving livelihood quality of the local community were identified and categorized into 6 classes.

Forest/ Shrub Land Conservation

10 ha Forest enrichment plantation

40 kilometer (km) Tree plantation along village road

Degraded Soil & Land Improvement

1 Landslide Treatment

15 Gully Treatment

Agriculture Land Conservation

70 ha Jhum to agroforestry

30 ha Fruit tree plantation

100 ha Orchard management

3 Drainage improvement

Stream Bank Protection

<1 km Buffer strip plantation and bio-engineering methods

Water Resources Development

0 Irrigation canal improvement

0 Spring development

Capacity Development & Demonstration

30 Households (HHs) Regenerative farming training

30 HHs Drip irrigation training

30 HHs Homestead improvement

Watershed Management Infrastructures

Protection Wall Construction	2
Earthen Dam Construction	9
Dam Improvement	3
Irrigation Canal Construction	5
Check Dam Construction	1

Watershed conservation interventions in Gilachhari include dam construction (9) and dam improvement (3) being the most implemented, followed by irrigation canal construction (5) aimed at enhancing water storage and distribution. In addition to these, other watershed conservation measures have been identified to further boost water availability, agricultural productivity, and overall ecosystem health. Measures such as check dam construction, culvert construction, and protection wall construction will help control runoff, reduce soil erosion, and protect critical infrastructure. Together, these interventions will significantly benefit the local community by ensuring year-round water access, improving crop yields, and reducing vulnerability to climate-induced hazards. These integrated efforts promote sustainable livelihoods and strengthen resilience in the Gilachhari region.

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